

Dynamic Monitoring of Radiological Pollution through Deep Learning

Classifying anomaly shapes in noisy gamma-dose time series

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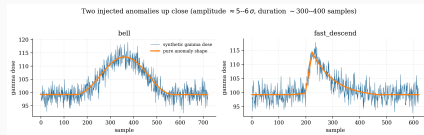
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Context: monitoring radiological pollution

- Environmental beacons count the **gamma photons** reaching a scintillation detector, once per minute → a **gamma-dose** time series.
- It reflects the natural radioactive background, modulated by weather and — rarely — by a radiological **event** (passing source, leak, plume).
- Operational goal: **detect and identify** such events as early as possible, from the *shape* they leave in the signal.

Why it is hard:

- per-sample noise $\approx$ the size of weak events;
- the baseline drifts slowly with weather $\Rightarrow$ no fixed threshold works;



Two events in the synthetic signal: a smooth **bell** (left) and an asymmetric **fast_descend** (right).

Problem and objectives

Related work and positioning

Signal model and synthetic data

Methods: architectures and training

Results

Conclusion and perspectives

Problem: windowed multi-class classification

Treat event identification as supervised classification of a univariate signal.

- Gamma-dose series $x = (x_1, \dots, x_N) \in \mathbb{R}^N$; x_t = value at minute t , N samples.
- Each instant belongs to one of K classes: label 0 = Normal Background, and $K - 1$ anomaly morphologies. Here $K = 7$.
- A class is defined by the local *shape*, not by $x_t \Rightarrow$ classify **fixed-length windows**, not samples.

Windowing and classifier

$$x^{(t)} = (x_{t-W+1}, \dots, x_t) \in \mathbb{R}^W, \quad \hat{p}^{(t)} = \text{softmax}(f_\theta(x^{(t)})) \in \Delta^{K-1}.$$

W = window length ($W = 512 \approx 8.5$ h); $f_\theta : \mathbb{R}^W \rightarrow \mathbb{R}^K$ = network with parameters θ (outputs class logits); Δ^{K-1} = probability simplex (entries ≥ 0 , summing to 1). Each window gets one label $y^{(t)} \in \{0, \dots, K - 1\}$ (coverage rule, later).

Objectives and approach

Training objective — class-weighted cross-entropy:

$$\mathcal{L}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} w_y \log \text{softmax}(f_\theta(x))_y, \quad w_c = \frac{|\mathcal{D}|}{K n_c}.$$

$\mathcal{D}$ = labelled training windows, $|\mathcal{D}|$ its size; n_c = number of windows of class c ; the weight w_c compensates class imbalance.

- **Evaluation objective:** window accuracy over-counts background $\Rightarrow$ report an **event-level macro F_1** : window probabilities are accumulated into a per-sample trace, contiguous non-background runs form predicted events, each is matched to a true event (precision/recall, F_1 averaged over anomaly classes).
- **Why deep learning:** a classical baseline-removal + matched-filter bank scales poorly (time-warp, noise, every new morphology $\Rightarrow$ redesign); CNN/attention *learn* shape features.
- **What we validate:** no labelled real events exist $\Rightarrow$ we train & evaluate on a

Related work — what we build on

Deep learning for time-series classification (TSC)

Shift from distance-based methods (1-NN + DTW) to deep nets. Ismail Fawaz et al. (2019) and Wang et al. (2017) find fully-convolutional and 1D residual networks the strongest general baselines → motivates our *conv baseline* and *residual* variant. TCNs (Bai et al. 2018): convolutions match/beat RNNs, more cheaply.

Convolutional / residual networks and normalisation

Weight-shared local filters (LeCun et al. 1998) give translation equivariance and length-independent parameters; residual connections (He et al. 2016) make depth trainable. BatchNorm (Ioffe & Szegedy 2015) vs GroupNorm (Wu & He 2018, identical at train/inference — exploited for a drifting amplitude).

Attention, hybrids, anomaly detection & explainability

Self-attention / Transformers (Vaswani et al. 2017), LayerNorm (Ba et al. 2016) and the [CLS] token (Devlin et al. 2019). CNN–Transformer hybrid for noisy physiological signals (Hu & Chen 2022, in the brief). We borrow event-level evaluation from the anomaly-detection survey (Blázquez-García et al. 2021); explainability via Grad-CAM (Selvaraju et al. 2017).

Positioning and contribution

- We **adopt established architectures** rather than proposing a new one: a convolutional baseline, a residual network (He 2016; Wang 2017), and a CNN–Transformer hybrid (Vaswani 2017; Hu 2022).

The contribution is methodological and applicative

1. a **simplified yet statistically calibrated** synthetic generator for noisy gamma-dose signals with controllable anomaly morphologies;
 2. a controlled, **like-for-like comparison** of the three architectures along the three axes named in the project brief — *classification performance*, *computational cost*, and *robustness to distribution shift*.
- To our knowledge this combination has not been reported for radiological-beacon anomaly classification.

Real data: a brief characterisation

- Four months of 2015 gamma-dose recordings (Feb, Apr, Jun, Oct), $\approx 40,000$ samples each, 1 / minute.
- Two components:
 - a slowly drifting **baseline** near 99 counts;
 - a zero-mean **high-frequency** fluctuation, $\sigma \approx 2.3\text{--}2.7$ counts.
- These few numbers are *all* we take from the real data — they calibrate the synthetic generator.



Synthetic background model

White Gaussian high-frequency noise on a slowly drifting Ornstein–Uhlenbeck (mean-reverting) baseline:

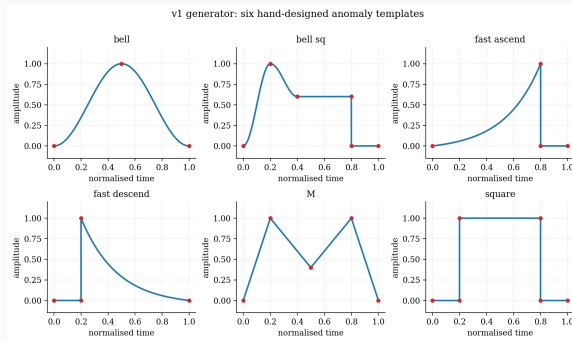
$$x_t = b_t + \eta_t, \quad \eta_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad b_{t+1} = b_t + \theta(\mu - b_t) + \xi_t, \quad \xi_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{\text{OU}}^2).$$

b_t = baseline; η_t = high-frequency sensor noise; μ = long-run mean level; θ = mean-reversion rate; σ = HF noise std; σ_{OU} = baseline-innovation std.

- Calibrated to the four real months: $\mu = 99.26$, $\sigma = 2.53$, $\theta = 4 \cdot 10^{-6}$, $\sigma_{\text{OU}} = 0.0373$.
- Tiny $\theta \Rightarrow$ the baseline drifts over hundreds of thousands of samples, far slower than the $W = 512$ window.
- **Deliberately crude**: it captures the marginal noise level and slow drift, but treats the HF noise as time-independent — the main simplification we accept (see Perspectives).

Anomaly morphologies

- Six morphologies $\rightarrow K = 7$ classes with the background: bell, bell_sq, fast_ascend, fast_descend, M, square.
- Stored as **keypoints** joined by linear / exponential / half-cosine segments.
- Every injected anomaly is **randomly deformed**: keypoints jittered by a Gaussian of std ν (the deformation level), shape-preserving constraints, then rescaled.

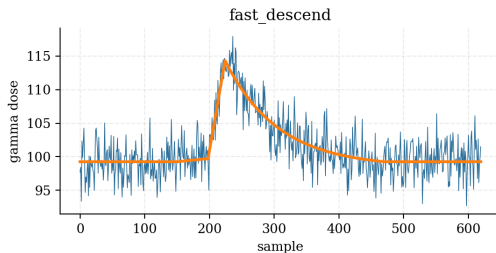
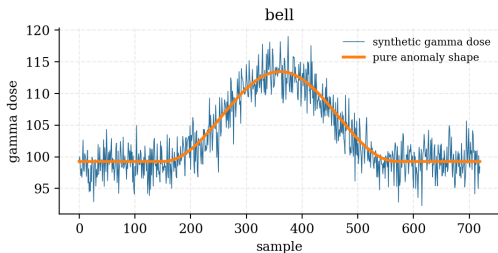


Sampling ranges

amplitude $\in \sigma \cdot [2.5, 7.6]$ (in noise- σ , i.e.

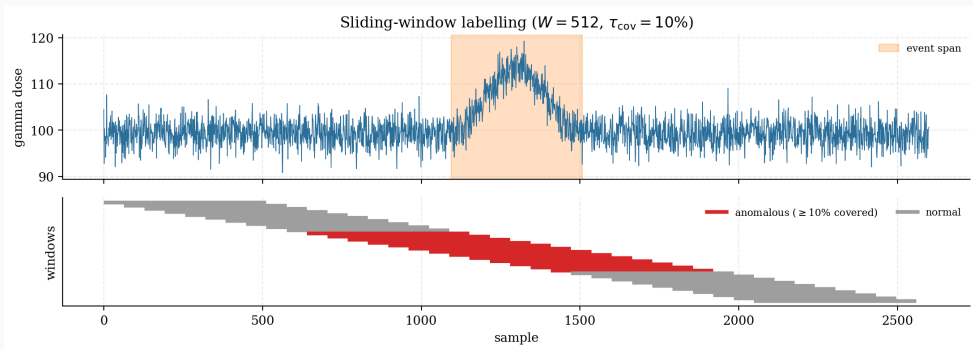
The synthetic dataset

Two injected anomalies up close (amplitude $\approx 5\text{--}6\sigma$, duration $\sim 300\text{--}400$ samples)



- Generator combines the background with injected, deformed anomalies (blue = generated dose, orange = clean shape before noise).
- Dataset used throughout: $N = 10^7$ samples, 5,000–8,000 anomalies ($\sim 6\text{--}8$ events / 10k samples).
- **Dense injection** = abundant, balanced examples of every morphology — at the cost of a background prior far denser than a real beacon's (a gap we revisit).

Input pipeline: windowing, labelling, normalisation



- **Temporal split first** (first 80% train+val, last 20% test), *then* window $\Rightarrow$ no leakage between overlapping windows. $W = 512$, stride 32 at training.
- **Coverage labelling**: a window is anomalous (majority anomaly class) iff $\geq \tau_{\text{cov}} = 10\%$ of its samples fall inside an event; otherwise **Normal**.
- **Per-window normalisation**: $\tilde{x}^{(t)} = (x^{(t)} - \overline{x^{(t)}}) / \sigma_{\text{train}}$, with $\sigma_{\text{train}} = 4.21$ (a single global std, reused at inference). The network sees only deviations from the local baseline.

Building blocks — convolution, normalisation, residual

1D convolution

$$v_{c,t} = b_c + \sum_{c'=1}^{C_{\text{in}}} \sum_{i=1}^k w_{c,c',i} u_{c',s\,t+i-p}.$$

$u \in \mathbb{R}^{C_{\text{in}} \times L}$ input (C_{in} channels, length L); $w \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k}$ kernel (width k); s stride; p padding; b_c bias; v output. Weight sharing $\Rightarrow$ translation equivariance; parameters $C_{\text{out}} C_{\text{in}} k$ *independent of* L . Then $\text{ReLU}(z) = \max(0, z)$, max-pooling, and `AdaptiveAvgPool1d(4)` (keeps 4 temporal bins).

Normalisation / Residual block

BatchNorm standardises each channel over the mini-batch (running stats differ train vs inference); **GroupNorm** standardises over channel *groups per sample* — identical at train/inference, amplitude-invariant. We use BN in the baseline, GroupNorm(8) in the two deeper models.

$$\text{Res}(u) = \text{ReLU}(u + \mathcal{F}(u)), \quad \mathcal{F} = \text{Conv1D-GN-ReLU-Conv1D-GN},$$

with a 1×1 conv on the shortcut when the channel count changes. The identity shortcut keeps deep stacks trainable (no vanishing gradient).

Building blocks — self-attention

Token sequence $Z \in \mathbb{R}^{T \times d}$: T tokens (time steps), dimension d .

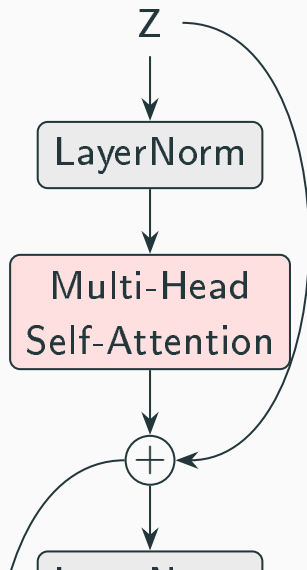
$$Q = ZW^Q, K = ZW^K, V = ZW^V,$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right)V.$$

$W^Q, W^K, W^V \in \mathbb{R}^{d \times d_h}$ learnable (d_h = per-head dimension).

Rows of $\text{softmax}(\cdot) \in \mathbb{R}^{T \times T}$ sum to 1 $\Rightarrow$ each output is a **convex combination of the value vectors** (learnt, content-based weighting).

- One attention layer lets *every* position see the whole window (unlike a bounded convolution).
- h heads in parallel (multi-head); a fixed **sinusoidal positional encoding** is added (attention is order-agnostic); a learnable [CLS] token is read out for classification.



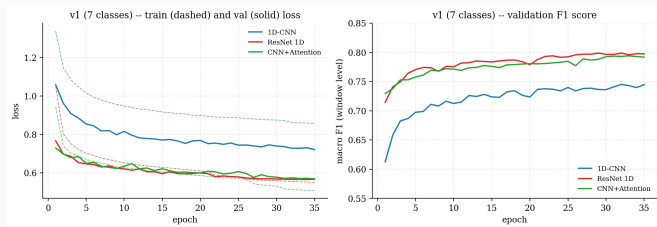
Three architectures of increasing complexity

Shared interface: input a single-channel window $x \in \mathbb{R}^{1 \times 512}$, output $K = 7$ logits.

	1D-CNN	ResNet 1D	CNN+Attention
Trunk	3 plain conv blocks	3 residual blocks	stride-2 stem + 2 encoders
Normalisation	BatchNorm	GroupNorm(8)	GroupNorm(8) + LayerNorm
Long-range mixing	—	—	multi-head self-attention
Parameters	22,727	74,631	109,655

- **A — 1D-CNN (baseline):** $3 \times \text{Conv-BN-ReLU-MaxPool}$ (kernels 7, 7, 5; channels $16 \rightarrow 32 \rightarrow 64$) + MLP head. Sanity check on the pipeline.
- **B — ResNet 1D:** same depth, each block residual with GroupNorm; *identical head* $\Rightarrow$ a clean ablation of the residual trunk.
- **C — CNN+Attention:** stride-2 conv stem $\rightarrow T = 64$ tokens of dimension $d = 64$, then 2 pre-norm Transformer encoders ($h = 4$ heads); [CLS] output feeds a linear classifier.

Training dynamics and window-level accuracy



Train/val loss (left) and validation macro F_1 (right).

- All converge within 35 epochs.
- The two deeper models reach a lower val loss (≈ 0.57 vs ≈ 0.72) and a higher val F_1 .

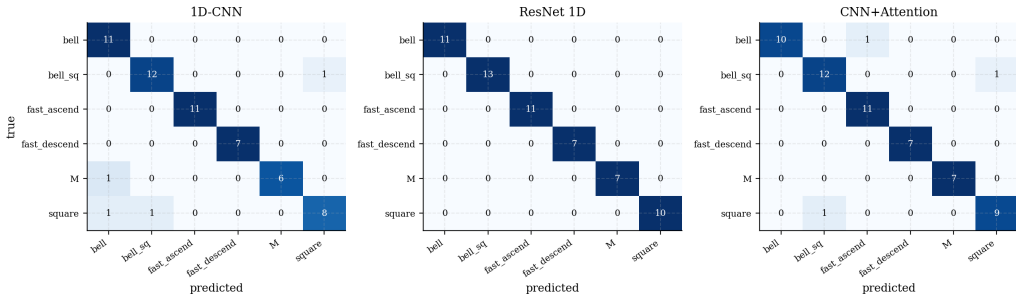
Overall window acc.

1D-CNN	0.766
ResNet 1D	0.807
CNN+Attention	0.795

Background $\sim 94\%$ for all; `bell_sq` is the hardest (a bell with a plateau). Depth helps.

Event-level evaluation (59-event synthetic segment)

v1 event-level confusion matrices (synthetic test segment, 59 events)

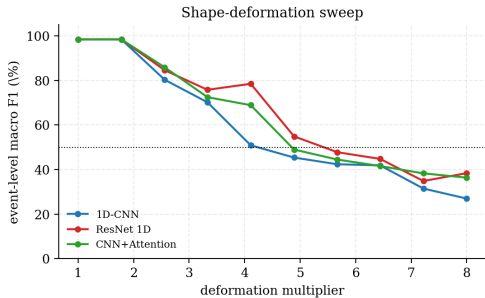
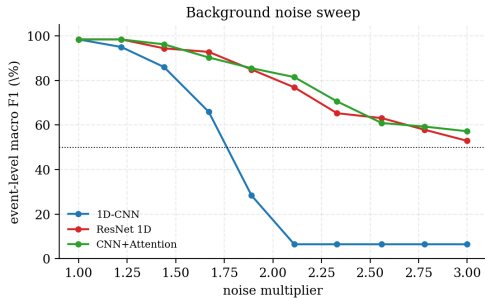


Model	Hits	Miss	FA	Macro F_1
1D-CNN	55	4	0	0.934
ResNet 1D	59	0	0	1.000
CNN+ Attention	56	3	0	0.955

- Events matched by interval overlap (FA = false alarm).
- **No false alarms** on the segment; the ResNet 1D classifies **all 59 events**; the other two miss visually similar morphologies.

Robustness to distribution shift

v1 robustness study (six classes, white-noise generator)



- Two axes (small regenerated test sets): background-noise multiplier $m_{\text{noise}} \in [1, 3]$ (noise *texture*, amplitude scaled with it) and shape-deformation multiplier $m_{\text{deform}} \in [1, 8]$. All start near 98%.
- **Noise:** the 1D-CNN drops below 50% by $m_{\text{noise}} \approx 1.9$ then **collapses to a single class ($\sim 6\%$)**; ResNet 1D & CNN+Attention stay $\geq 53\text{--}57\%$ to $3\times$. **Deformation:** all degrade gracefully ($< 50\%$ at 4.9 / 5.7 / 4.9).
- **GroupNorm + residual** buy the noise robustness; the baseline's batch-norm running statistics

Computational cost and discussion

Model	Parameters	Train time	Inference (win/s)	Event F_1
1D-CNN	22,727	~ 4.3 min	~ 10,700	0.934
ResNet 1D	74,631	~ 10.0 min	~ 14,200	1.000
CNN+Attention	109,655	~ 10.2 min	~ 15,200	0.955

Table 1: *

Single RTX 5060 Laptop GPU.

- **Residual learning + GroupNorm is the biggest win:** at equal depth and identical head, ResNet 1D beats the plain 1D-CNN on every axis, and the gap explodes under noise.
- **Self-attention buys no extra robustness here:** $W = 512$ with events ≤ 800 samples leaves little long-range context for attention to exploit.
- **The cheapest model is the most fragile:** the 1D-CNN's single-class collapse makes it unsuitable across seasons.
- $\Rightarrow$ [ResNet 1D 1D](#) is the [best trade-off](#) of accuracy, robustness and cost.

Conclusion and recommendation

- Formalised radiological-event identification as [sliding-window multi-class classification](#) of a noisy gamma-dose signal, and validated a deep-learning approach on a controlled, calibrated synthetic benchmark.
- All three architectures reach strong event-level performance (macro F_1 from 0.93 to 1.00, no false alarms).
- The ResNet 1D is best overall — perfect event score, highest window accuracy, and (with the hybrid) the only model that resists background-noise growth, while the baseline collapses. Attention did *not* help at the 512-sample window scale.

Recommendation

ResNet 1D 1D — best accuracy and robustness at moderate cost ($\sim 75\text{k}$ params, ~ 10 min on a laptop GPU). The 1D-CNN only if compute is very constrained *and* noise is stable; the CNN+Attention if windows are later extended to capture long-range context.

Limitations

- Results are, by construction, [synthetic](#).
- HF noise is independent in time (real sensor noise is correlated).
- Anomaly density far higher than a real beacon's.
- No labelled real test set $\Rightarrow$ real performance not yet measurable.

Perspectives — “to go further”

- [Realistic signal models](#): coloured (autoregressive) noise matching the real autocorrelation; lower density; calibrated morphologies — *same pipeline* re-run.
- [A labelled real test set](#): first real event-level precision / recall.
- [Explainability](#): Grad-CAM on conv feature maps; attention maps.
- [Longer windows & richer inputs](#): meteorological channels; the context attention lacked here.

Thank you — questions?

On a clean, controlled benchmark the approach works; *depth and group normalisation*,
not attention, buy robustness at this window length.

Backup — shared training / inference settings

Setting	Value
Window W / stride (train / inf.)	512 / 32 / 10
Labelling threshold τ_{cov}	0.10 (majority anomaly class)
Normalisation	per-window centring, $\sigma_{\text{train}} = 4.21$
Split	temporal 80/20, then random 80/20 on train
Optimiser	Adam ($\beta_1=0.9, \beta_2=0.999, \epsilon=10^{-8}$)
Learning rate / batch	10^{-3} / 128, max 35 epochs
LR scheduler / early stopping	ReduceLROnPlateau(2, 0.5) / patience 10
Loss / class weights w	weighted CE; (0.46, 1.18, 1.27, 1.32, 1.24, 1.15, 1.31)
Train / test windows	249,985 / 62,485
Event extraction	bg-prob. threshold 0.3, bridge gaps ≤ 50

Backup — synthetic generator configuration

Parameter	Value
Total length N	10^7 samples (written in $5 \cdot 10^4$ chunks)
Number of anomalies	5,000–8,000 ($\sim 6\text{--}8$ / 10k samples)
Injection jitter	± 200 samples
Anomaly amplitude	$\sigma \cdot \mathcal{U}(2.5, 7.6)$ (SNR units)
Anomaly duration	$\mathcal{U}[200, 800]$ samples
Deformation level ν	$\mathcal{U}(0.02, 0.10)$
HF noise	i.i.d. $\mathcal{N}(0, \sigma^2)$, $\sigma = 2.53$
Baseline (OU)	$\mu = 99.26$, $\theta = 4 \cdot 10^{-6}$, $\sigma_{\text{OU}} = 0.0373$
Templates / classes	6 morphologies + background ($K = 7$)

Backup — key references

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